



Research Paper

Short Variable Regions *flaA* Gene (SVR-*flaA*) Diversity and Virulence Profile of Multidrug – Resistant *Campylobacter* from Poultry and Poultry Meat in India



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ABSTRACT

Human gastrointestinal infections caused by *Campylobacter* species is the second most important foodborne illness after salmonellosis worldwide. Poultry represent one of the main sources of *Campylobacter* organisms. In the present study, the short variable region of flagellin gene (SVR-*flaA*) typing was carried out to determine the variation among the circulating strains of *Campylobacter jejuni* and *Campylobacter coli*. The *C. jejuni* and *C. coli* isolated from poultry and poultry meat were screened for the presence of virulence determinants like *cadF*, *flaA*, *cdtB*, and *wlaN* gene. The screening for *wlaN* gene is crucial in view of the fact that most patients with Guillain Barre's (GB) syndrome with a preceding history of diarrheal illness have been found to harbor *wlaN* gene – positive *C. jejuni* strains. Out of the 200 samples comprising poultry meat and cloacal swabs, 21.5% of samples were found to harbor *Campylobacter* spp. of which 2.5% were *Campylobacter jejuni*, and 19% were confirmed as *Campylobacter coli*. The *cadF*, *flaA*, *cdtB* virulence genes were detected in all the *Campylobacter* spp. isolated in the present study. The presence of the *wlaN* gene in the *Campylobacter jejuni* isolated in the present study may pose a public health threat with long – term human health implications. The SVR-*flaA* typing of twelve *Campylobacter* isolates obtained in the present study revealed that *Campylobacter coli flaA* sequence OL471375 is a new strain with a novel allele type 1,675 and peptide sequence 5 which stands deposited in pubMLST database for *Campylobacter*. The other *flaA*-SVR gene sequences identified in this study were OL471369, OL471370, OL471371, OL471372, OL471373, and OL471374. Among twelve *Campylobacter* spp., three distinct *DdeI*-RFLP patterns were observed, each varying in size from 100 to 1,000 base pairs. Antimicrobial profiling of the *Campylobacter* spp. isolated in the present study revealed that 50% of the strains were multidrug resistant. All the *Campylobacter* spp. were resistant to ciprofloxacin (CIP), ampicillin (AMP), penicillin (PEN), and nalidixic acid (NAL) whereas 57.1% of strains were resistant to tetracycline (TET) and erythromycin (ERY) 28% to amoxicillin (AMX) and enrofloxacin (ENO), 85% to amikacin (AMK). The high degree of resistance to fluoroquinolones observed in the present study is crucial in view of fluoroquinolones being drugs of choice for the treatment of human *Campylobacter* infections.

Campylobacter is a leading foodborne bacterial pathogen linked with the consumption of food producing animals worldwide (Li et al., 2019). *Campylobacter* infections caused by consumption of contaminated poultry meat and products represent about 50–70% of the global *Campylobacter* infection burden (Seliwiorstow et al., 2015). It is the most common cause of bacterial gastroenteritis in developed

and developing countries responsible for 400–500 million cases of diarrhea worldwide each year. In the USA, more than 1% population acquires this infection each year (Ruiz-Palacios, 2007). Although diarrhea is the most common clinical manifestation of infection in humans, 30% of cases with a preceding history of *Campylobacter* infection may develop Guillain Barre syndrome (GBS), an acute demyelinating

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disease of the peripheral nervous system, within 1–3 weeks after the onset of diarrheal illness (Mangen et al., 2015).

The molecular mimicry hypothesis between glycosylated lipopoligosaccharides and gangliosides on neuronal cells results in the onset of GBS. Other sequelae like Miller-Fisher syndrome, Reiter's syndrome, and irritable bowel syndrome have also been reported following *Campylobacter* gastroenteritis (Mangen et al., 2015).

Campylobacter jejuni and *Campylobacter coli* are the two main species frequently associated with human disease accounting for more than 95% of human infections (Kvalsvig et al., 2014). In Europe, food-borne campylobacteriosis is most frequently a notified disease since 2005, whereas in the USA, it is the second most occurring bacterial illness in humans preceded by the infection due to *Salmonella* spp. During a two-year study conducted at a tertiary care hospital in the Union Territory of Jammu and Kashmir India, out of 92 patients presenting with different types of GBS, 71 had a preceding history of bacterial gastroenteritis and/or pregnancy. Cases were observed to increase during the spring and summer months (Wani et al., 2022). Although scarce data from low- and middle-income countries still suggest an increasing rate of *Campylobacter* infection in humans. Despite preventive measures like risk assessment along the “farm-to-table,” and improved understanding of its pathogenesis, *Campylobacter* infections continue to show an upward trend globally.

The highly pathogenic human *Campylobacter* species include *Campylobacter jejuni* subsp. *jejuni* (Cjj), *Campylobacter jejuni* subsp. *doyley* (Cjd), *Campylobacter coli* and the mildly pathogenic ones are *Campylobacter concisus*, *Campylobacter upsaliensis*, *Campylobacter lari*, and *Campylobacter hyointestinalis*. The veterinary pathogens include *Campylobacter fetus* subsp. *venerealis* (Cfv) and *Campylobacter fetus* subsp. *fetus* (Cff) (Igwaran and Okoh, 2019). Among them, *C. jejuni* is the major culprit of human infections, causing the vast majority of cases compared to other *Campylobacter* species (Epps et al., 2013).

Poultry are the largest reservoir of *Campylobacter* spp. and the droppings of birds contain 10^5 to 10^8 CFU/g *Campylobacter* organisms (Epps et al., 2013). Experimental studies in poultry have revealed that the infected birds excrete *C. jejuni*, from the second or third week (Acuff et al., 1982) of infection and develop weight loss, gut damage, and impaired intestinal permeability (Awad et al., 2018). Potential sources of entry of the organism in the flock include old litter, untreated drinking water, other farm animals, domestic pets, wildlife species, insects, equipment transport vehicles, and farm workers. The wild birds (Passeriformes, Columbiformes, and Anseriformes) in nonbird proof poultry farms may also be the source of transmission of *C. jejuni* to poultry.

The frequent inter- and intraspecies genetic recombination leads to genetic diversity among *C. jejuni* isolates. Several molecular subtyping methods with enhanced discriminatory power have been used to investigate the epidemiology of *C. jejuni* (Wieczorek et al., 2017). The SVR-*flaA* typing method has been reported to be a useful and convenient cost-effective sequence-based tool employed in epidemiological studies involving a small number of isolates (Moore et al., 2005). The next-generation sequencing, whole – genome sequencing methods possess increased discriminatory power over other methods. However, such methods are not cost – effective in low-income countries. The sequencing of PCR-amplified short variable regions (SVRs) of the flagellin-encoding (*flaA*) gene located between 450 and 600 bp is a helpful and economical tool for *Campylobacter* genotyping (Harrington et al., 1997). The *flaA*-SVR nucleotide allele sequences are available in the pubMLST *Campylobacter* database, with open access to all the *flaA*-SVR types isolated worldwide. The SVR-*flaA* typing provides sufficient discrimination for its use as a subtyping method for *C. jejuni* due to the fact that some of the regions of the flagellin encoding A gene are genetically stable over a long period thereby helping in molecular typing and differentiation of isolates (Manning et al., 2003).

Fluoroquinolones, macrolides, third and fourth cephalosporins, glycopeptides, and polymyxins have been categorized by WHO as “highest priority critically important” antibiotics in human medicine for the treatment of bacterial infections in view of the limited availability of alternatives. Garcia-Migura et al. (2014) highlighted a concerning trend in their study. They found a direct link between indiscriminate use of fluoroquinolone use in poultry farming and the rising incidence of multidrug – resistant *Campylobacter* strains in humans underscoring the urgent need for more judicious use of antibiotics in the poultry industry.

The pubMLST *Campylobacter* database currently lacks information on SVR-*flaA* types, virulence determinants, and strains of *Campylobacter* found in poultry, poultry meat, or products in India. This is a significant concern because the Union Territory (UT) of Jammu and Kashmir (J&K) contributes a major portion to the country's poultry consumption. Given the relatively unorganized farm-to-table practices in the poultry sector in J&K, there could be a potential risk to the local population here thereby emphasizing the need for more comprehensive data on *Campylobacter* in the region. The goal of this study is to investigate the prevalence of *Campylobacter* spp. in both poultry and poultry meat, to understand the virulence profile, to identify the types of *flaA* present, and to evaluate the potential risk to humans. This is particularly relevant given reports of patients with Guillain-Barré syndrome in hospitals within the union territory.

Materials and methods

Sample collection

The isolation of *Campylobacter* spp. from the poultry and poultry meat was carried out as described previously (Wieczorek et al. (2018)). Briefly, a total of 200 samples comprising 86 cloacal swabs, 50 enteritis samples from Government poultry farms (Hariparbat and Broiler farm at Faculty of Veterinary Sciences & Animal Husbandry SKUAST-K), and 64 fresh broiler meat samples (poultry wings and drumsticks) from local retail vendors around the Srinagar and the Ganderbal districts of Kashmir valley were collected. The broiler meat samples were collected in inverted plastic bags, and prepackaged samples (plastic wrapped, modified atmosphere, or vacuum packaged) were taken directly from refrigerated or freezer cabinets, from supermarkets and butcher shops. The swabs and enteritis samples were collected in sterile vials in Cary Blairs transport medium (Hi Media, India) and transported to the laboratory on ice and stored at 4 °C till further processing. All samples were tested within 48–72 hrs of collection.

Sample processing, bacterial strain isolation, phenotypic characterization, and molecular typing

The primary enrichment of the samples was carried under microaerophilic conditions at 42 °C for 24–48 hrs in Campy-thioglycollate broth (Hi-Media, India) supplemented with Blaser-Wang supplement (Hi-Media, India). To ensure no *Campylobacter* colonies were overlooked, primary enrichment isolation was performed using two types of selective media. Each enriched sample was streaked on *Campylobacter* agar base medium (Hi-Media, India) containing 10% defibrinated sheep blood, supplemented with Blaser-Wang (Hi-Media, India), and on modified Cefoperazone Charcoal Agar (mCCDA) (Hi-Media, India) supplemented with CCDA supplement (Hi-Media, India). The plates were incubated under microaerophilic conditions (campygen gas pack, Oxoid) at 42 °C for 24–48hrs. Characteristic mucoid, nonhemolytic colonies on *Campylobacter* agar base medium containing 10% sheep blood with irregular edges were screened for further identification. On mCCDA grey, flat, swarming colonies with oil drop appearance were screened for further confirmatory tests. The primary

identification of suspected colonies was based on colony morphology, gram staining, and biochemical tests (catalase, oxidase, and Hippurate hydrolysis test). The study was planned to determine the occurrence of *Campylobacter* spp in both poultry and poultry meat, determine the virulence profile of the local strains, identify the *flaA* types, and evaluate the potential human risk based on the virulence profile of the strains isolated in the study.

Bacterial DNA extraction & PCR assays

DNA was extracted using the Wizard DNA extraction kit (Promega). The isolates were confirmed by genus-specific primer (16SrRNA) of *Campylobacter* spp. and multiplex PCR using species-specific primers of *C. jejuni* and *C. coli* as per the protocol of Denis et al. (1999). All the PCR assays in the present study were performed in 25 µl reaction volume in the Mastercycler gradient (Eppendorf, Germany). The reaction mixture comprised of 3.0 µl template DNA, 12.5 µl of GoTaq® Green Master Mix, 0.5 µM concentration of each primer, and sterile nuclease-free water (NFW) to make up the volume. Sterile water was incorporated as the negative control. The *C. jejuni* and *C. coli*–positive controls were kindly provided by Prof. Linda Van der Graff, Department of Infectious Disease and Immunology, Faculty of Veterinary Medicine, Utrecht University, Netherland. The cycling conditions comprised of initial denaturation at 94 °C for 5 min, followed by 35 cycles of denaturation at 94 °C for 1 min, annealing specific to the primer pair for 60sec, and extension at 72 °C for 60 sec, followed by a final extension at 72 °C for 10 min. The virulence gene profile of *C. jejuni* and *C. coli* for *cadF*, *cdtB*, and *wlaN* genes was determined. The primer, annealing temperatures, and genes of the study are depicted in Table 1. The amplified PCR products were visualized in 1.5% agarose gel in tris acetate EDTA (TAE) buffer, 0.5 µg/ml ethidium bromide (EtBr) and visualized under ultraviolet illumination and documented using Gel Documentation System (UHV, Imaging System, Thermo fisher).

Short variable region-*flaA* (SVR *flaA*) typing

Short variable region (SVR) typing of *flaA* gene of twelve *Campylobacter* isolates in the present study was done using the primer sequences listed in Table 1. The cycling conditions for *flaA* typing consisted of initial denaturation at 94 °C for 4 min, followed by 30 cycles of denaturation at 94 °C–30 sec, annealing at 46 °C–30 sec, and extension at 72 °C–30 sec, followed by a final extension at 72 °C for 5 min and visualization in 1.5% agarose gel. The PCR products were purified using the QIA quick PCR Purification Kit (Qiagen). The cloning of the products was done by ligating to pGEM-T Easy Vector using a PCR cloning kit (pGEM®-T Easy Vector Systems, Promega). The ligation reaction mixture (10 µl) comprised of 5 µl of 2X rapid ligation buffer,

1 µl of pGEMT Easy Vector (50 ng), 1 µl T4 DNA Ligase (3 Weiss units/µl), 2 µl of purified PCR product, and sterile nuclease-free water. The ligation reaction mixture was incubated for 1 h at room temperature. The *Escherichia coli* DH5α competent cells were prepared using 0.1 M of CaCl₂ solution. The transformation was carried out by adding 2.5 µl of ligation mixture to 100 µl of competent cells. A volume of 100 µl of transformed cells was pipetted out and poured on LB agar plates containing ampicillin (100 µg/ml), X-gal (40 mg/ml), and IPTG (100 mM). The plates were incubated at 37 °C overnight. The white – colored colonies were screened for the presence of insert using colony PCR. The plasmids were extracted using a plasmid extraction kit (Pure-Yield™ Plasmid Miniprep System). The sanger sequencing of cloned plasmids was done at AgriGenome Labs Pvt. Ltd, (Kerala, India) using. The *flaA* sequences were submitted to Genbank, and one novel peptide sequence of the present study was deposited in pubMLST database for *Campylobacter* (https://pubmlst.org/pubmlstcampylobacter_isolates). The evolutionary history was inferred by using the Maximum Likelihood method based on the Tamura-Nei model. Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Maximum Composite Likelihood (MCL) approach and then selecting the topology with superior log likelihood value. The tree was drawn to scale, with branch lengths measured in the number of substitutions per site. The analysis involved 7 nucleotide sequences. A total of 402 positions were identified in the final dataset. Evolutionary analyses were conducted in MEGA X. Representative SVR *flaA* sequence types were deposited in GenBank.

Restriction fragment length polymorphism (RFLP)

The full – length *flaA* gene (1,728 bp) of *Campylobacter* was amplified using the primer sequence given in Table 1. The cyclic conditions used comprised of initial denaturation at 94 °C for 1 min, followed by 35 cycles of denaturation at 92 °C–30 sec, annealing at 52 °C–1.5 min and extension at 72 °C–2.5 min, and final extension at 72 °C–5 min. The amplified *flaA* gene products (1,728 bp) were digested using HpyF3I: *DdeI* (ThermoScientific™) restriction enzyme as per the protocol of Nachamkin et al. (1999). The digestion mixture consisted of 1 µl of 4U *DdeI*, 2 µl of 10X Restriction buffer, 10 µl of PCR product, and the final volume was made with nuclease – free water and incubated at 37 °C for 16 hrs, and the products were resolved on 2% agarose gel.

Antimicrobial resistance study

The antimicrobial susceptibility testing was also performed using the Kirby Bauer disc diffusion test as per CLSI guidelines 2021 (Clinical Laboratory Standards Institute, Performance Standards for Antimicro-

Table 1
Primer sequences used in present study

Target	Primer sequence (5'–3')	Annealing (°C)	Product size (bp)	Reference
16S rRNA (Genus specific)	F-GGTTAAGTCCCGCAACGAGCCGC R-GGCTGATCTACGATTACTAGCGAT	50	283	Leite et al. (1996)
<i>mapA</i> (<i>C.jejuni</i>)	F-CTATTTTATTTTGTAGTGCTTG R-GCTTTATTTGCCATTGTTTATTA	52	589	Denis et al. (1999)
<i>ceuE</i> (<i>C.coli</i>)	F-AATTGAAAATTGCTCCAACATATG R-TGATTTTATTATTTGTAGCAGCG	52	462	–do–
<i>cdtB</i>	F-CAGAAAGCAAATGGAGTGT R-AGCTAAAGCGGTGGAGTAT	52	620	Hickey et al. (2000)
<i>cadF</i>	F-TTGAAGGTAATTTAGATATG R-CTAATACCTAAAGTTGAAAC	45	400	Konkel et al. (1999)
<i>wlaN</i>	F-TTAAGAGCAAGATATGAAGGTG R-CCATTTGAATTGATATTTTG	46	672	Linton et al. (2000)
SVR- <i>flaA</i>	F-5'CTATGGATGAGCAATTAATAA3' R-5'CAAGTCCTGTTCCAAGTGAAG3'	46	402	Berrang et al. (2003)
<i>flaA</i>	F-5'GGATTTCGTATTAACACAAATGGTGC3 R-5'CTGTAGTAATCTTAAACATTTTG-3'	52	1,728	Nachamkin et al. (1999)

bial Susceptibility Testing). The panel of antibiotic discs included penicillin (10mcg), amoxicillin (10 µg), ampicillin-sulbactam (10/10 mcg), ampicillin (10mcg), amikacin(30mcg), tetracycline(30 µg), ciprofloxacin(5mcg), enrofloxacin (5mcg), and erythromycin (15mcg) and nalidixic acid(30 µg). A fresh 18–24 h broth culture of *Campylobacter* strains corresponding 0.5 McFarland opacity tube was diluted 1:10 in sterile normal saline and spread by swab on Mueller–Hinton agar containing 5% fresh sheep blood (Hi Media India), followed by application of antibiotic discs (Hi Media, India). The plates were incubated in a microaerophilic atmosphere at 37 °C for 24 h. Multiple drug resistance (MDR), which is based on resistance shown by a species of microorganism to at least one antimicrobial drug in three or more antimicrobial categories, was used to categorize isolates of *Campylobacter* as MDR.

Results

The occurrence of *Campylobacter* in poultry and poultry meat was 21.5% of which *C. jejuni* was, 2.5% and *C. coli*, 19% (Fig. 1a). The broiler meat samples (poultry wings and drumsticks) carried the highest percentage (26.5%) of *Campylobacter* spp followed by cloacal swabs (24.4%) and enteric samples (10%) as detected by PCR. The proportion of *Campylobacter*-positive samples differed significantly among the live birds and meat ($P < 0.001$) samples. The cloacal swab samples from the broiler production unit of Hariparbat farm recorded a higher incidence (53.8%) of *Campylobacter* organisms in comparison to the poultry rearing unit of the same farm (50%) followed by FVSc & A.H poultry farm (13.11%). The retail broiler meat vendors from the Gandarbhal area had a higher incidence (36%) of *Campylobacter* spp. in broiler meat samples than those from the Srinagar area (20.5%). In the present study in total of twelve isolates, comprising three *Campylobacter jejuni* and nine *Campylobacter coli* strains were isolated.

Among the diverse virulence factors having a role in *Campylobacter* pathogenesis, *cdtB*, and *cadF* genes were detected in all strains. The *wlaN* virulence gene coding for β -1,3-galactosyltransferase enzyme that can elicit GBS in humans was detected in all *Campylobacter jejuni* (Fig. 1b). Of the seven (07) *flaA*-SVR(CAMP1255) types identified in the present study, one *flaA*-SVR sequence from “*C. coli*” sequence unique to this geographical region has been identified. The *flaA*-SVR sequence types identified from twelve *Campylobacter* strains in the present study are 62, 33, 78, 79, 72, and 44. The most common sequence type 33 was identified in five strains of *Campylobacter* spp. The representative *flaA*-SVR genotypes from different samples along with Gen-

bank accession numbers are depicted in Table 2. This new *flaA* sequence type of *C. coli* (OL471375) has not been reported previously in pubMLST *Campylobacter* and has been assigned *fla*-nucleotide allele 1,675, (peptide sequence 5). The analysis of *flaA* gene sequence in pubMLST *Campylobacter* database revealed that *C. coli* isolates JKP-A (OL471369), JKP-E (OL471373), JKP-F (OL471374), and JKP-G (OL471375) exhibited the highest homology (>95%) to allele type 62, 33, 33 and 78, respectively, whereas *C. jejuni* isolates JKP-B (OL471370), JKP-C (OL471371) and JKP-D (OL471372) had higher homology with allele types 79, 72 and 44, respectively, reported in pubMLST *Campylobacter*. In the present study, *flaA* *C. coli* sequence types OL471372 and OL471373 are more closely related to the OL471371 sequence than OL471374. The *flaA* type OL471369 is closely related to *flaA* sequence C25 and C44 of *C. coli* reported from other areas of the globe than to the *flaA* sequence OL471375 and OL471373 of the present study. On scrutinizing the *flaA* sequences of *Campylobacter coli* in the pubMLST *Campylobacter* database, it is evident that the sequence type OL471369 shares a close relationship with several other isolates of *Campylobacter* spp. These include the AR-0420 *Campylobacter coli*, the *flaA* of *Campylobacter jejuni* FDAARGOS_422, *C. jejuni* CFSAN032806, *C. jejuni* subsp. *jejuni* NCTC 11168-GSv, and the *flaA* gene of *C. jejuni* CS0048. This relationship is depicted in Figure 2. Among the *Campylobacter* isolates of this study the *flaA* gene, sequence of OL471375 and OL471373 of *C. coli* strains are more closely related to the OL471374 *flaA* gene, sequence of *C. jejuni* strain isolated in the present study.

DdeI-RFLP analysis of 1,728 bp full-length *flaA* gene of twelve strains of *Campylobacter* spp. revealed three different patterns with band sizes ranging from 100 to 1,000 bp, indicative of strain variation among the *Campylobacter* isolates (Fig. 3). However, the *DdeI*-RFLP pattern could detect only three different circulating strains in comparison to seven detected by *flaA*-SVR typing. In this study, all the examined *Campylobacter* isolates exhibited resistance to several antibiotics, namely ciprofloxacin (CIP), ampicillin (AMP), penicillin (PEN), and nalidixic acid (NAL). Notably, half of these isolates demonstrated multidrug resistance. Furthermore, 57.1% of the strains showed resistance to both tetracycline (TET) and erythromycin (ERY), while 28% were resistant to amoxicillin (AMX) and enrofloxacin (ENO). Alarmingly, a high proportion of strains (85%) were resistant to amikacin (AMK), an antibiotic often used as a last resort for severe infections in humans. These findings underscore the growing challenge of antibiotic resistance in *Campylobacter* spp with a potential of transmission of such resistant isolates to humans through the food chain.

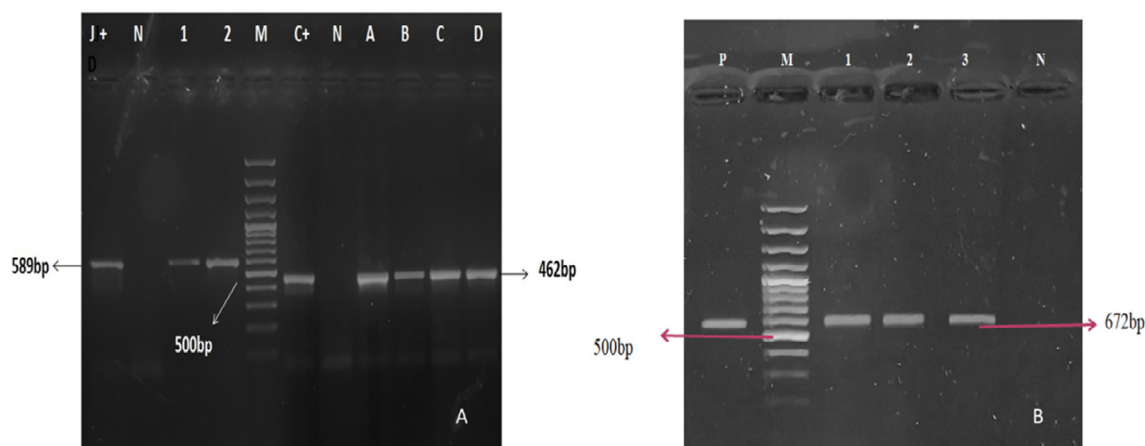


Figure 1. A. Multiplex PCR for *C. jejuni* and *C. coli*. J: Positive control *Campylobacter jejuni*, N: Negative control, 1–2: Isolates positive for *Campylobacter jejuni* (589 bp), M: 100 bp ladder, C: Positive control *Campylobacter coli*, N: Negative control, A–E: isolates positive for *Campylobacter coli* (462). **B. *wlaN* gene PCR for *C. jejuni*.** P: Positive control, M: 100 bp ladder, 1–3: Isolates positive for *wlaN* gene (672 bp), N: Negative control.

Table 2
SVR *flaA* types of *Campylobacter* spp. in poultry birds of Kashmir.

(Gen Bank Accession No)	Species	Source	Match Sequence Similarity type (>95%)
JKP-A (OL471369)	<i>C. coli</i>	Cloacal swab	CAMP1255(<i>flaA</i>):62
JKP-B (OL471370)	<i>C. jejuni</i>	Broiler meat	CAMP1255(<i>flaA</i>):79
JKP-C (OL471371)	<i>C. jejuni</i>	Broiler meat	CAMP1255(<i>flaA</i>):72
JKP-D (OL471372)	<i>C. jejuni</i>	Cloacal swab	CAMP1255(<i>flaA</i>):44
JKP-E (OL471373)	<i>C. coli</i>	Cloacal swab	CAMP1255(<i>flaA</i>):33
JKP-F (OL471374)	<i>C. coli</i>	Cloacal swab	CAMP1255(<i>flaA</i>):33
JKP-G (OL471375)	<i>C. coli</i>	Cloacal swab	CAMP1255(<i>flaA</i>):78

Discussion

The flagellin gene locus of *Campylobacter* is composed of two genes *flaA* and *flaB* (Khoshbakht et al., 2015). The short variable region (SVR) typing of the *flaA* gene of *Campylobacter* spp used in the present study has been used by workers for molecular typing of *Campylobacter* spp. The SVR *flaA* typing method of *Campylobacter* refers to the sequencing of PCR-amplified short variable regions (SVRs) of the flagellin-encoding (*flaA*) gene. This method is a helpful and economical tool for *Campylobacter* genotyping. It has been used to assess the molecular diversity of *Campylobacter jejuni* isolated from various sources and to analyze the relationship of the isolate genotypes with

their antimicrobial resistance profiles and source of isolation. In the context of *Campylobacter*, the pubMLST database stores the *flaA* SVR sequence types. In a study by Wieczorek et al. (2019) of multiresistant *Campylobacter jejuni*, it was found that 13 isolates had novel *flaA* SVR subtypes which were not present in the pubMLST database. These isolates were recovered from various sources including chicken feces, carcasses, meat, and humans.

The SVR *flaA* typing method of *Campylobacter* has discriminatory power with the sequence data and genotypic information stored publicly in an accessible pubMLST database. The SVR-*flaA* gene variability among the strains in the present study may be attributed to the high variability of the *Campylobacter* genome caused by its instability (Wieczorek et al., 2019). The *flaA* flagellar gene has been reported to undergo spontaneous mutations during host infections which may have a crucial role in molecular variation (Guerry, 2007). *flaA*-SVR type 33 of the pubMLST *Campylobacter* database was the most common sequence identified in the present study. *flaA*-SVR typing has been used to study the genetic diversity of *Campylobacter* isolates by other researchers (Gomes et al., 2016; Magnússon et al., 2011; Singh & Kwon, 2013; Wassenaar et al., 2009).

Out of the seven (07) *flaA*-SVR types identified, one sequence is unique to its origin. The sequences identified were 62, 33, 78, 79, 72, and 44. A similar genetic diversity in SVR region of the *flaA* gene of *Campylobacter* spp has been reported by several workers (Corcoran et al., 2006; Wassenaar et al., 2009; Wiecezorek et al., 2019). Corcoran

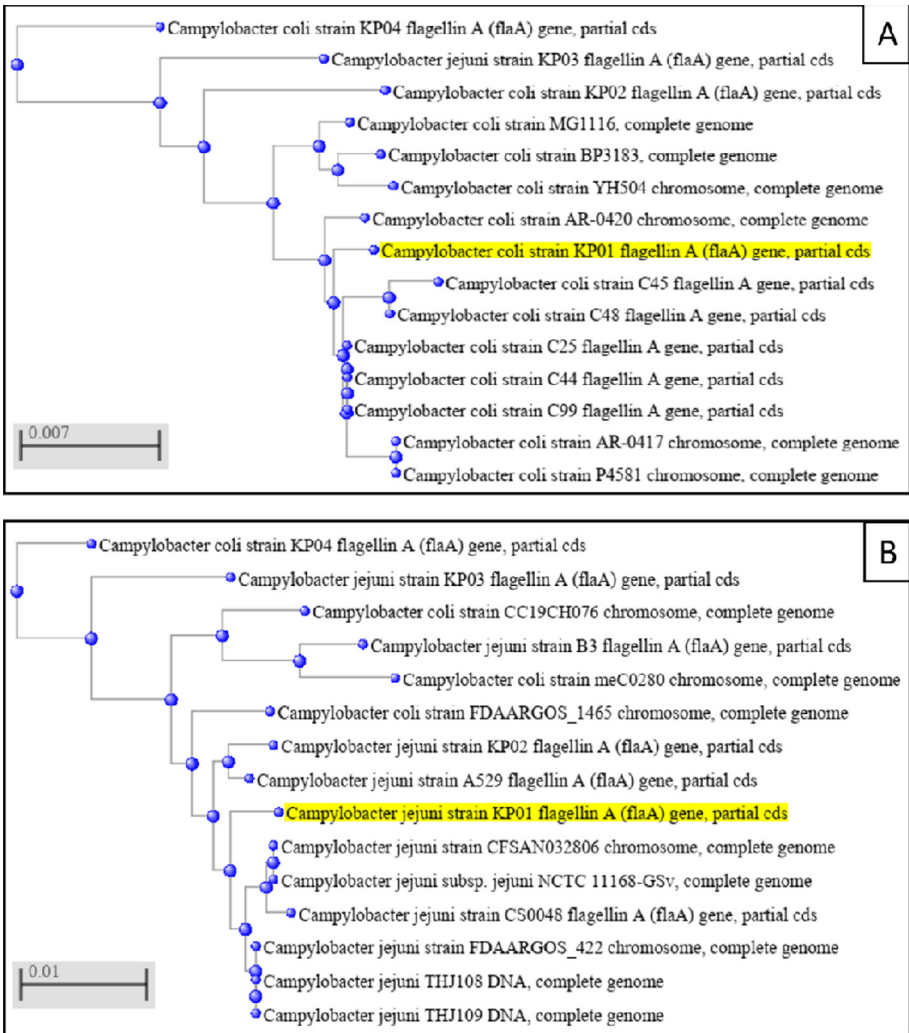


Figure. 2. A. Phylogenetic analysis of *Campylobacter coli*. **B.** Phylogenetic analysis of *Campylobacter jejuni*.

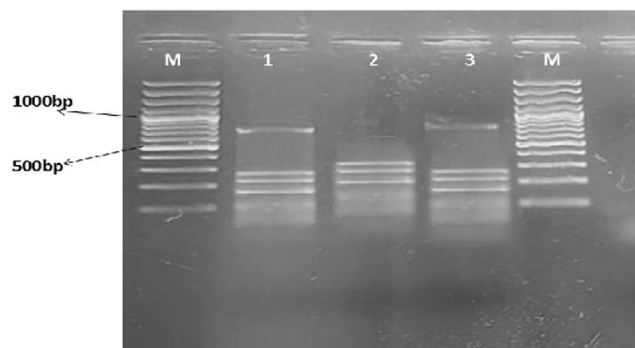


Figure 3. *DdeI*-RFLP digestion of *flaA* gene of *Campylobacter* isolates. M: 100 bp ladder, 1–3: Digestion pattern of amplified 1,728 bp *flaA* gene.

et al. (2006) reported SVR-*flaA* allele types 325, 66, 343, 78, 427, 8, and 410 in the poultry population of Ireland. Wieczorek et al. (2019) observed the presence of 16, 54, 36.287, 78, 14, 49, 100 SVR-*flaA* allele types in meat and feces of poultry in Poland. Wassenaar et al. (2009) reported a total of 67 allele types in poultry from three European regions. The common allele types found in poultry were 36, 8, 70, 21, 32, and 34.

The *flaA* gene has also been found suitable for restriction fragment length polymorphism (RFLP) analysis (Wassenaar & Newell, 2000). In view of good discriminatory power, the restriction enzyme *DdeI* has been preferred for typing *Campylobacter* spp. (Ghorbanalizadgan et al., 2016). The *DdeI*-RFLP pattern identified three different strains based on the banding pattern (100–1,000 bp). In this study, *flaA*-SVR typing identified seven different allele types among the isolates in comparison to *DdeI*-RFLP which could identify only three different strains. Yadav et al. (2018) reported 15 different *DdeI*-RFLP patterns in *Campylobacter* isolates ranging from 200 to 1,100 bp.

In the present study, 64 broiler meat samples were screened for the presence of *Campylobacter* spp., and 17(26.5%) were positive. Out of these, 3 (17.6%) were identified as *C. jejuni* and 14 (82.3%) as *C. coli*. Sinulingga et al. (2020) reported an occurrence of 68.9% (*C. jejuni*) and 17.2% (*C. coli*) from poultry meat samples collected from wet markets and supermarkets in Malaysia. The *cadF* gene product adhesin and fibronectin-binding proteins help *Campylobacter* to adhere to the extracellular matrix of intestinal cells (Khoshbakht et al., 2013). *Cdt* blocks G2/M phase of eukaryotic cells prior to cell division, inducing cytoplasmic distension ultimately leading to cell death (Reddy & Zishiri, 2013). Glycosyl transferase, a product of *wlaN*, glycosylates the LOS of *C. jejuni*. The glycosylated LOS mimics the structure of glycans and gangliosides receptors present in neuronal cells of humans. The immunoglobulins produced in response to the *C. jejuni* epitopes cross-react with human neurons resulting in Guillain-Barre syndrome (GBS) neuropathy. The presence of *wlaN* gene in all *C. jejuni* strains isolated in the present study is indicative of a potential human health hazard. Datta et al. (2013) reported the presence of this gene in 23.8% and 4.7% of broiler meat and broiler feces, respectively. Wieczorek et al. (2018) reported a 13.7% presence of the *wlaN* gene, in *C. jejuni* isolates from chicken feces and 12.5% from poultry meat. Since three *C. jejuni* strains were found to harbor the gene, *wlaN* gene cannot be a true representative of the distribution in poultry here. Nevertheless, its presence in *C. jejuni* strains in the present study is of concern in view of the zoonotic potential of this organism.

The research on the antimicrobial profile and genetic diversity of *Campylobacter jejuni* and *C. coli* isolates is crucial for managing and treating campylobacteriosis. The unregulated use of antibiotics in poultry farming has contributed to the rise of antibiotic-resistant *Campylobacter* spp. in humans. In the present study, half of the isolates demonstrated multidrug resistance. Over half of the strains (57.1%) were resistant to both tetracycline and erythromycin and 28% showed

resistance to amoxicillin and enrofloxacin. A high degree of resistance (85%) to amikacin, a critical antibiotic for severe human infections was also noted. These findings underscore the growing challenge of antibiotic resistance in *Campylobacter* spp with a potential of transmission of such resistant isolates to humans through the food chain. The lowered resistance to enrofloxacin (28%) in this comparison to other antibiotics has been previously reported earlier (Koenraadi et al., 1995). Gharbi et al. (2018) reported the highest resistance of *Campylobacter* isolates to tetracycline (100%) and erythromycin (100%) followed by ciprofloxacin (99.2%), ampicillin (61.4%) and aminoglycosides (12.9%) antibiotics.

In the present study, seven different *flaA*-SVR subtypes were identified. A novel allele (1,675) not reported previously in pubMLST *Campylobacter* database was identified in this study. In the present study, 50% of the strains were multidrug-resistant and all the strains were resistant to ciprofloxacin (fluoroquinolone). The increased fluoroquinolone resistance as well as an increase in number of MDR *Campylobacter* spp is concerning in view of fluoroquinolones being the drugs of choice for the treatment of Campylobacteriosis in humans. The increase in antimicrobial resistance calls for reducing antimicrobial use in poultry and incorporation of specific procedures to reduce chicken carcass contamination. The presence of *wlaN* gene coding for an enzyme that can elicit GBS in humans in *Campylobacter jejuni* isolates from chicken is also a concern in view of carry-over human infections from poultry meat and products.

Declaration of competing interests

The authors declared that they have no competing interests.

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CRediT authorship contribution statement

Saima Iqbal: Methodology. **Sabia Qureshi:** Supervision, Project administration, Conceptualization. **Muddasir S. Bandy:** Formal analysis. **Shaheen Farooq:** Investigation, Methodology. **Zahid A. Kashoo:** Investigation, Supervision. **Maliha Gulzar:** Methodology, Manuscript writing. **M. Altaf Bhat:** Formal analysis, Project administration. **Arif Pandit:** Formal analysis. **Md. Isfaquul Hussain:** Validation. **Pervaiz Dar:** Writing – review & editing. **Gulzar Badroo:** Writing – review & editing. **Mahrugh Hafiz:** Methodology. **Faheem ud Din:** Data curation. **Junaid Mehraj:** Formal analysis.

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References

- Acuff, G. R., Vanderzant, C., Gardner, F. A., & Golan, F. A. (1982). Examination of turkey eggs, poult and broiler house facilities for *Campylobacter jejuni*. *Journal of Food Protection*, 45, 1279–1281.
- Awad, W. A., Hess, C., & Hess, M. (2018). Re-thinking the chicken–*Campylobacter jejuni* interaction: a review. *Avian Pathology*, 47, 352–363.
- Berrang, M. E., Meinersmann, R. J., Buhr, R. J., Reimer, N. A., Philips, R. W., & Harrison, M. A. (2003). Presence of *Campylobacter* in the respiratory tract of broiler carcasses before and after commercial scalding. *Poultry Science*, 82, 1995–1999.
- Corcoran, D., Quinn, T., Cotter, L., Whyte, P., & Fanning, S. (2006). Antimicrobial resistance profiling and fla-typing of Irish thermophilic *Campylobacter* spp. of human and poultry origin. *Letters in Applied Microbiology*, 43, 560–565. <https://doi.org/10.1111/j.1472-765X.2006.01987.x>.

- CLSI Performance Standards for Antimicrobial Susceptibility Testing (2021). *CLSI supplement M100* (31st ed.). Clinical and Laboratory Standards Institute.
- Datta, S., Niwa, H., & Itoh, K. (2013). Prevalence of 11 pathogenic genes of *Campylobacter jejuni* by PCR in strains isolated from humans, poultry meat and broiler and bovine faeces. *Journal Medical Microbiology*, 52, 345–348.
- Denis, M., Soumet, C., Rivoal, K., Ermel, G., Blivet, D., Salvat, G., & Colin, P. (1999). Development of a m-PCR assay for simultaneous identification of *Campylobacter jejuni* and *C. coli*. *Letters in Applied Microbiology*, 29, 406–410.
- Epps, S. V., Harvey, R. B., Hume, M. E., Phillips, T. D., Anderson, R. C., & Nisbet, D. J. (2013). Foodborne *Campylobacter* spp. infections, metabolism, pathogenesis and reservoirs. *International Journal of Environmental Research and Public Health*, 10, 6292–6304.
- Garcia-Migura, L., Hendriksen, R. S., Fraile, L., & Aarestrup, F. M. (2014). Antimicrobial resistance of zoonotic and commensal bacteria in Europe: the missing link between consumption and resistance in veterinary medicine. *Veterinary Microbiology*, 170, 1–9.
- Gharbi, M., Béjaoui, A., Hamda, C. B., Kais, A. J., Zrelli, G. C., Hamrouni, S., Aouadhi, C., Bessoussa, G. A., & Maaroufi, A. (2018). Prevalence and Antibiotic Resistance Patterns of *Campylobacter* spp. Isolated from broiler chickens in the north of Tunisia. *Biomedical Research International* 943786.
- Ghorbanalizadgan, M., Bakhshi, B., & Najar-Peerayeh, S. (2016). Application of PCR-RFLP provides as discrimination as total flaA sequence analysis among *C. jejuni* isolates of clinical origin. *Japanese Journal of Infectious Diseases*, 69, 373–377. <https://doi.org/10.7883/yoken.JJID.2015.386>.
- Gomes, C. N., Souza, R. A., Passaglia, J., Duque, S. S., Medeiros, M. I., & Falcao, J. P. (2016). Genotyping of *Campylobacter coli* strains isolated in Brazil suggests possible contamination amongst environmental, human, animal and food sources. *Journal of Medical Microbiology*, 65, 80–90. <https://doi.org/10.1099/jmm.0.000201>.
- Guerry, P. (2007). *Campylobacter* flagella: not just for motility. *Trends in Microbiology*, 15, 456–461. <https://doi.org/10.1016/j.tim.2007.09.006>.
- Harrington, C. S. F., Thomson-Carter, M., & Carter, P. E. (1997). Evidence for recombination in the flagellin locus of *Campylobacter jejuni*: implications for the flagellin gene typing scheme. *Journal of Clinical Microbiology*, 35, 2386–2392.
- Hickey, T. E., McVeigh, A. L., Scott, D. A., Michielutti, R. E., Bixby, A., Carroll, S. A., Bourgeois, A. L., & Guerry, P. (2000). *Campylobacter jejuni* cytolethal distending toxin mediates release of interleukin-8 from intestinal epithelial cells. *Infection and Immunology*, 68, 6535–6541.
- Igwaran, A., & Okoh, A. I. (2019). Human *Campylobacteriosis*: a public health concern of global importance. *Heliyon*, 5, e02814.
- Kvalsvig, M. G., Baker, A., & Sears, N. F. (2014). Bacteria: campylobacter. *Encyclopedia of Food Safety*, 1, 369–380.
- Khoshbakht, R., Tabatabaei, M., Hosseinzadeh, S., Aski, H. S., & Seifi, S. (2015). Genetic characterization of *Campylobacter jejuni* and *C. coli* isolated from broilers using flaA PCR-restriction fragment length polymorphism method in Shiraz, Southern Iran. *Jundishapur Journal Microbiology*, 8, e18573.
- Khoshbakht, R., Tabatabaei, M., Hosseinzadeh, S., Shekarforoush, S. S., & Aski, H. S. (2013). Distribution of nine virulence-associated genes in *Campylobacter jejuni* and *C. coli* isolated from broiler feces in Shiraz, Southern Iran. *Foodborne Pathogenic Disease*, 10, 764–770.
- Koenraadi, P. M. F. J., Jacobs-Reitsma, W. F., Van der Laan, T., Beumer, R. R., & Rombouts, F. M. (1995). Antibiotic susceptibility of *Campylobacter* isolates from sewage and poultry abattoir drain water. *Epidemiology and Infection*, 115, 475–483.
- Konkel, M. E., Gray, S. A., Kim, B. J., Garvis, S. G., & Yoon, J. (1999). Identification of the enteropathogens *Campylobacter jejuni* and *Campylobacter coli* based on the *cadF* virulence gene and its product. *Journal of Clinical Microbiology*, 37, 510–517.
- Leite, P., Blom, K., Patton, C. M., Nicholson, M. A., Steigerwalt, A. G., Hunter, S. B., Brenner, S. B., Barrett, T. J., & Bala, S. (1996). Rapid identification of *Campylobacter* species by restriction fragment length polymorphism analysis of a PCR-amplified fragment of the gene coding for 16S rRNA. *Journal of Clinical Microbiology*, 34, 62–67.
- Li, M., Havelaar, A. H., Hoffmann, S., Hald, T., Kirk, M. D., Torgerson, P. R., & Devleeschauwer, B. (2019). Global disease burden of pathogens in animal source foods, 2010. *PLoS one*, 14, e0216545.
- Linton, D., Gilbert, M., Hitchen, P. G., Dell, A., Morris, R. H., Wakarchuk, W. W., Gregson, N. A., & Wren, B. W. (2000). Phase variation of a β 1,3 galactosyltransferase involved in generation of the ganglioside GM₁ like lipo oligosaccharide of *Campylobacter jejuni*. *Molecular Microbiology*, 37, 501–514.
- Magnússon, S. H., Guðmundsdóttir, S., Reynisson, E., Rúnarsson, A. R., Harðardóttir, H., Gunnarson, E., Georgsson, F., Reiersen, J., & Marteinsson, V. Th. (2011). Comparison of *Campylobacter jejuni* isolates from human, food, veterinary and environmental sources in Iceland using PFGE, MLST and fla-SVR sequencing. *Journal of Applied Microbiology*, 111, 971–981.
- Mangen, M. J. J., Bouwknegt, M., Friesema, I. H., Haagsma, J. A., Kortbeek, L. M., Tariq, L., Wilson, M., Van Pelt, W., & Havelaar, A. H. (2015). Cost-of-illness and disease burden of food-related pathogens in the Netherlands. *International Journal of Food Microbiology*, 196, 84–93.
- Manning, G., Dowson, C. G., Bagnall, M. C., Ahmed, I. H., West, M., & Newell, D. G. (2003). Multilocus sequence typing for comparison of veterinary and human isolates of *Campylobacter jejuni*. *Applied Environmental Microbiology*, 69, 6370–6379.
- Moore, J., Corcoran, D., Dooley, J., Fanning, S., Lucey, B., Matsuda, M., McDowell, D., Megraud, F., Cherie Millar, B., O'Mahony, R., Riordan, L., Rourke, M., Rao, J. R., Rooney, P. J., Sails, A., & Whyte, P. (2005). *Campylobacter*. *Veterinary Research*, 36, 351–382.
- Nachamkin, I., Bohachick, K., & Patton, C. M. (1999). Flagellin gene typing of *Campylobacter jejuni* by restriction fragment length polymorphism analysis. *Journal of Clinical Microbiology*, 37, 1531–1536.
- Reddy, S., & Zishiri, O. T. (2013). Genetic characterisation of virulence genes associated with adherence, invasion and cytotoxicity in *Campylobacter* spp. isolated from commercial chickens and human clinical cases. *Orderstepoort Journal of Veterinary Research*, 85, 1–9.
- Ruiz-Palacios, G. M. (2007). The health burden of *Campylobacter* infection and the impact of antimicrobial resistance: playing chicken. *Clinical Infection and Disease*, 44, 701–703.
- Seliwiorstow, T., Duarte, A., Baré, J., Botteldoorn, N., Dierick, K., Uyttendaele, M., & De Zutter, L. (2015). Comparison of sample types and analytical methods for the detection of highly *Campylobacter*-colonized broiler flocks at different stages in the poultry meat production chain. *Foodborne Pathogens and Disease*, 12, 399–405.
- Singh, P., & Kwon, Y. M. (2013). Comparative analysis of *Campylobacter* populations within individual market-age broilers using fla gene typing method. *Poultry Science*, 92, 2135–2144. <https://doi.org/10.3382/ps.2012-02615>.
- Sinulingga, T. S., Aziz, S. A., Bitrus, A. A., Zunita, Z., & Abu, J. (2020). Occurrence of *Campylobacter* species from broiler chickens and chicken meat in Malaysia. *Tropical Animal Health Production*, 52, 151–157.
- Wani, Y. I., Hassan, Z., Baba, T. H., Farooq, S., & Mehraj, I. (2022). Pattern of GBS in Kashmir, a Northern Region in India. *European Journal of Molecular and Clinical Medicine*, 9, 1218–1225.
- Wassenaar, T. M., Fernández-Astorga, A., Alonso, R., Marteinsson, V. T., Magnússon, S. H., Kristoffersen, A. B., & Hofshagen, M. (2009). Comparison of *Campylobacter* fla-SVR genotypes isolated from humans and poultry in three European regions. *Letters in Applied Microbiology*, 49, 388–395.
- Wassenaar, T. M., & Newell, D. G. (2000). Genotyping of *Campylobacter* spp.. *Applied Environmental Microbiology*, 66, 1–9.
- Wieczorek, K., Wołkiewicz, T., & Osek, J. (2019). FlaA-SVR Based genetic diversity of multiresistant *Campylobacter jejuni* isolated from chickens and humans. *Frontiers in Microbiology*, 10, 1176.
- Wieczorek, K., Wołkiewicz, T., & Osek, J. (2018). Antimicrobial resistance and virulence-associated traits of *Campylobacter jejuni* isolated from poultry food chain and humans with diarrhea. *Frontiers in Microbiology*, 9, 1508.
- Wieczorek, K., Denis, E., Lachtara, B., & Osek, J. (2017). Distribution of *Campylobacter jejuni* multilocus sequence types isolated from chickens in Poland. *Poultry Science*, 96, 703–709.
- Yadav, R., Yadav, J., Maherchandani, S., & Kashyap, S. K. (2018). Typing of *Campylobacter jejuni* isolated from poultry on the basis of flaA-RFLP by various restriction enzymes. *Veterinary and Animal Science*, 6, 1–5.